

BUSINESS REQUIREMENTS DOCUMENT

DairyPro 360 — All-in-One Dairy Management System

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Table of Contents

1. Executive Summary
2. Business Context & Background
3. Problem Statement
4. Stakeholders
5. Business Objectives & Success Metrics
6. Project Scope
7. High-Level Business Requirements
 - 7.1 Farm & Cattle Management
 - 7.2 Milk Collection & Quality Testing
 - 7.3 Processing & Production
 - 7.4 Inventory Management
 - 7.5 Distribution & Delivery Management
 - 7.6 Sales, Billing & Finance
 - 7.7 HR & Payroll
 - 7.8 Reports & Analytics
 - 7.9 User Management & Security
8. Non-Functional Requirements
9. Integration Requirements
10. Compliance & Regulatory Requirements
11. Assumptions & Constraints
12. Risks
13. Glossary

1. Executive Summary

DairyPro 360 is a comprehensive, cloud-enabled Dairy Management System designed to digitise and streamline every operational layer of an all-in-one dairy enterprise — from cattle farming and raw milk collection through to processing, packaging, distribution, and retail. The system replaces fragmented, paper-based or siloed tools with a single integrated platform accessible by multiple roles across the organisation.

This Business Requirements Document (BRD) captures the high-level business needs, stakeholder expectations, functional scope, and constraints that will guide the design, development, and deployment of DairyPro 360. It serves as the contractual baseline between the business and the development team.

2. Business Context & Background

The dairy industry is one of the most time-sensitive and compliance-heavy sectors in agri-food business. A mid-to-large dairy operation today manages hundreds of cattle, processes thousands of litres of milk daily, employs dozens of field agents, and delivers across a multi-tier distribution network. Current operational challenges include:

- Manual milk collection registers that are error-prone and hard to audit.
- No real-time visibility into cattle health, yield, or feed consumption.
- Disconnected invoicing and payment systems causing revenue leakage.
- Inability to trace a quality defect back to source (farm, batch, route).
- HR and payroll managed in spreadsheets, leading to disputes.
- No demand forecasting, causing over/under production losses.

DairyPro 360 addresses all of the above by providing a single source of truth for operations, finance, quality, and logistics.

3. Problem Statement

The business currently operates with an estimated **12–18% operational inefficiency** due to the absence of an integrated management system. Key pain points are:

Pain Point	Business Impact	Severity
Manual milk ledger entries	Data errors, disputes with farmers	High
No cattle health tracking	Undetected disease spread, yield drop	High
Paper-based quality tests	No traceability, failed audits	High
Disconnected billing	Delayed payments, revenue leakage	High
No route optimisation	High fuel costs, late deliveries	Medium
Spreadsheet HR/payroll	Salary disputes, compliance risk	Medium
No analytics dashboard	Reactive decisions, missed trends	Medium

4. Stakeholders

Stakeholder	Role	Interest
Dairy Owner/MD	Primary Sponsor	ROI, compliance, strategic dashboard
Farm Manager	Key User	Cattle records, feed, health alerts
Milk Collection Agent	Key User	Collection entry, fat/SNF testing
Processing Supervisor	Key User	Batch production, QC, equipment logs
Distribution Manager	Key User	Route planning, delivery tracking
Finance Officer	Key User	Billing, payments, ledger, P&L
HR Manager	Key User	Employee records, payroll, attendance
Quality Analyst	Key User	Lab results, compliance reports
Retailers/Customers	External	Order placement, delivery status
Farmers (Suppliers)	External	Milk supply data, payment statements
Regulatory Bodies	Compliance	FSSAI, PFA Act, GST compliance
IT Administrator	System Admin	User access, system uptime, backups

5. Business Objectives & Success Metrics

#	Objective	KPI / Success Metric
BO-01	Eliminate paper-based collection logs	100% digital entries within 3 months
BO-02	Improve milk quality traceability	Full batch-to-farm trace in < 2 mins
BO-03	Reduce billing disputes	Zero manual billing errors after go-live
BO-04	Enable real-time cattle health tracking	Alert generated within 1 hr of anomaly
BO-05	Optimise delivery routes	15% reduction in fuel costs within 6 months
BO-06	Centralise HR & payroll	Payroll processed in < 2 hours monthly
BO-07	Provide analytics for decision-making	Executive dashboard available 24/7
BO-08	Ensure regulatory compliance	Zero FSSAI non-compliance incidents

6. Project Scope

6.1 In Scope

- Farm and cattle lifecycle management (birth, health, breeding, disposal)
- Raw milk collection recording with fat/SNF/CLR testing
- Processing and production batch management
- Inventory management (raw materials, finished goods, packaging)
- Distribution and delivery tracking with route management
- Sales, invoicing, payment collection, and financial reporting
- HR management: employee records, attendance, leave, payroll
- Quality control: lab testing, FSSAI compliance logs
- Role-based user management and access control
- Mobile-responsive web application
- Dashboards and analytical reports
- SMS/Email alert notifications
- Integration with weighing scales and milk analysers (via API)

6.2 Out of Scope

- Custom ERP integrations (SAP/Oracle) — Phase 2
- IoT sensor hardware procurement and installation
- Third-party logistics partner portals
- Retail POS system
- Livestock insurance management

7. High-Level Business Requirements

7.1 Farm & Cattle Management

- BR-FM-01: The system shall maintain a digital register of all cattle (ID, breed, age, health status, vaccination records).
- BR-FM-02: The system shall record daily feed consumption per cattle and alert on deviations.
- BR-FM-03: The system shall track milk yield per cattle per session and flag underperforming animals.
- BR-FM-04: The system shall schedule and log veterinary visits, vaccinations, and treatments.
- BR-FM-05: The system shall record breeding events and pregnancy tracking.
- BR-FM-06: The system shall generate cattle performance reports monthly.

7.2 Milk Collection & Quality Testing

- BR-MC-01: The system shall record raw milk collection per farmer/supplier per shift (morning/evening).
- BR-MC-02: The system shall capture fat %, SNF %, CLR, and temperature readings for each collection.
- BR-MC-03: The system shall calculate payment due per farmer based on quantity × rate × fat content.
- BR-MC-04: The system shall flag milk that fails quality thresholds and prevent it from entering processing.
- BR-MC-05: The system shall generate collection summary reports daily, weekly, and monthly.
- BR-MC-06: The system shall support bulk data import from milk analysers via CSV or API.

7.3 Processing & Production

- BR-PP-01: The system shall manage production batches with input quantities, output quantities, and by-products.
- BR-PP-02: The system shall track product categories: pasteurised milk, curd, butter, ghee, cheese, paneer.
- BR-PP-03: The system shall record equipment usage, downtime, and maintenance schedules.
- BR-PP-04: The system shall enforce FIFO (First In First Out) for raw material usage.
- BR-PP-05: The system shall provide real-time production status to managers.
- BR-PP-06: The system shall calculate production yield efficiency per batch.

7.4 Inventory Management

- BR-IN-01: The system shall track raw material stock (milk, packaging, additives) with reorder alerts.
- BR-IN-02: The system shall track finished goods inventory by SKU, batch, and expiry date.
- BR-IN-03: The system shall deduct inventory automatically on dispatch and add on production completion.
- BR-IN-04: The system shall support multiple warehouses/cold storage units.
- BR-IN-05: The system shall generate stock valuation reports at any point in time.

7.5 Distribution & Delivery Management

- BR-DD-01: The system shall manage delivery routes with assigned vehicles and drivers.
- BR-DD-02: The system shall generate daily delivery schedules based on orders.

- BR-DD-03: The system shall track delivery status (dispatched, in-transit, delivered, returned).
- BR-DD-04: The system shall record proof of delivery (digital sign-off or OTP confirmation).
- BR-DD-05: The system shall calculate and log vehicle fuel consumption and trip costs.
- BR-DD-06: The system shall handle return/rejection cases with reason codes.

7.6 Sales, Billing & Finance

- BR-SF-01: The system shall manage customer (retailer/institutional) profiles and credit limits.
- BR-SF-02: The system shall generate GST-compliant invoices automatically on order dispatch.
- BR-SF-03: The system shall track accounts receivable and send payment reminders.
- BR-SF-04: The system shall record farmer payments (accounts payable) linked to collection records.
- BR-SF-05: The system shall produce a Profit & Loss summary by product category.
- BR-SF-06: The system shall integrate with a payment gateway for online collections.

7.7 HR & Payroll

- BR-HR-01: The system shall maintain employee master data (permanent and contract).
- BR-HR-02: The system shall support biometric/OTP-based attendance marking.
- BR-HR-03: The system shall calculate monthly payroll with deductions (PF, ESI, TDS).
- BR-HR-04: The system shall manage leave requests and balances.
- BR-HR-05: The system shall generate payslips and Form 16 summaries.

7.8 Reports & Analytics

- BR-RA-01: The system shall provide an executive dashboard with KPIs: milk volume, revenue, cattle count.
- BR-RA-02: The system shall support trend analysis for production, sales, and quality over time.
- BR-RA-03: The system shall allow custom date-range reports exportable as PDF/Excel.
- BR-RA-04: The system shall provide farmer-wise contribution reports.
- BR-RA-05: The system shall include demand forecasting charts based on historical sales.

7.9 User Management & Security

- BR-US-01: The system shall support role-based access control (RBAC) with pre-defined roles.
- BR-US-02: The system shall enforce password policies and two-factor authentication for admins.
- BR-US-03: The system shall log all user actions (audit trail) with timestamp and IP.
- BR-US-04: The system shall allow the admin to create, deactivate, and assign roles to users.
- BR-US-05: The system shall mask sensitive financial data from non-finance roles.

8. Non-Functional Requirements

Category	Requirement	Target
Performance	Page load time	< 2 seconds (95th percentile)
Performance	API response time	< 500 ms
Availability	System uptime	99.5% monthly SLA
Scalability	Concurrent users	Up to 500 simultaneous users
Security	Data encryption	AES-256 at rest, TLS 1.3 in transit
Security	Authentication	JWT + optional 2FA
Usability	Mobile responsiveness	Fully functional on mobile browsers
Usability	UI language	English + Hindi (Phase 1)
Backup	Data backup	Daily automated backup, 30-day retention
Disaster Recovery	RTO / RPO	RTO 4 hrs / RPO 1 hr
Compliance	FSSAI data logs	Immutable audit trail, 5-year retention
Compliance	GST invoicing	Compliant with GST Council norms

9. Integration Requirements

- Milk Analyser Machines: REST API or CSV import for fat/SNF readings.
- Weighing Scales: Serial/USB or API integration for automated weight capture.
- Payment Gateway: Razorpay / PayU for online customer payment collection.
- SMS Gateway: Twilio / MSG91 for collection alerts, payment reminders, and notifications.
- Email Server: SMTP (SendGrid or similar) for automated reports and alerts.
- Government Portals: GSTIN API for invoice validation (Phase 2).
- Accounting Software: Tally XML export for finance reconciliation (Phase 2).

10. Compliance & Regulatory Requirements

- FSSAI (Food Safety and Standards Authority of India): Maintain production batch logs, quality test records, and supplier traceability as required for dairy licence.
- Prevention of Food Adulteration (PFA) Act: Quality parameters must be recorded and traceable per batch.
- GST Compliance: All invoices must carry GSTIN, HSN codes for dairy products, and correct tax slabs.
- Labour Laws: Payroll must comply with PF, ESI, and Minimum Wages Act.
- Data Protection: Customer and employee data must be stored securely per the IT Act 2000 and DPDP Act 2023.

11. Assumptions & Constraints

11.1 Assumptions

- Stable internet connectivity will be available at all collection centres.
- The client will provide master data (cattle IDs, farmer names, SKU lists) before go-live.
- Hardware (weighing scales, milk analysers) will be procured by the client.
- The initial deployment will cover one dairy unit; additional units in Phase 2.
- Users will undergo a 2-day training programme before going live.
- Regulatory rates (GST, PF, ESI) are accurate as of development and may need updates.

11.2 Constraints

- Project budget: INR 25–40 lakhs (estimated).
- Go-live target: 6 months from BRD sign-off.
- Technology stack must be open-source-friendly to minimise licensing costs.
- System must function on low-end Android devices (2 GB RAM) for field agents.
- All documents and UI labels must be available in English.

12. Risks

Risk ID	Description	Likelihood	Impact	Mitigation
R-01	User resistance to digital adoption	High	High	Training + change management plan
R-02	Data migration errors from manual records	Medium	High	Dry-run migration + validation scripts
R-03	Internet outage at collection centres	Medium	High	Offline mode with sync on reconnect
R-04	Hardware integration delays	Medium	Medium	Use manual entry as fallback
R-05	Scope creep during development	High	Medium	Strict change control process
R-06	Regulatory changes (GST rates)	Low	Medium	Configurable tax tables in admin panel

13. Glossary

Term	Definition
BRD	Business Requirements Document — high-level business needs
FRD	Functional Requirements Document — detailed functional specs
SNF	Solids-Not-Fat — quality parameter for milk
CLR	Corrected Lactometer Reading — density indicator of milk
FAT%	Percentage of fat content in milk
FSSAI	Food Safety and Standards Authority of India
FIFO	First In First Out — inventory management principle
SKU	Stock Keeping Unit — unique product identifier
RBAC	Role-Based Access Control

GST	Goods and Services Tax
PF/ESI	Provident Fund / Employees' State Insurance
KPI	Key Performance Indicator
RTO/RPO	Recovery Time/Point Objective — disaster recovery metrics
JWT	JSON Web Token — authentication mechanism